

# Microwave Imaging Framework

## Full Modules Overview Report

What each module does, status, and where to see outputs

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Modules completed: 12 / 12 (core) | Unit tests: 95 / 95 passing

## 1. Executive summary

This framework implements an end-to-end microwave breast imaging pipeline: load validated measurement data, preprocess S21 (especially for UG Touchstone briefs), reconstruct images with DAS/DMAS/DMAS-D4, localize and refine a suspicious ROI, decide tumor candidacy, characterize the region, score reconstruction confidence, visualize results, and export a session report. All twelve planned modules are implemented at functional/core level.

## 2. End-to-end flow

| Step                 | Modules     | User action / output                           |
|----------------------|-------------|------------------------------------------------|
| Load                 | 1           | Choose .mat / BMID scan / .s2p                 |
| Preprocess           | 2 (+3 path) | BMID pass-through or .s2p Week 1–3 pipeline    |
| Reconstruct          | 3–5         | DAS/DMAS/DMAS-D4 + auto/manual beamformer pick |
| Localize & detect    | 6–8         | ROI, high-res refine, tumor candidate Yes/No   |
| Characterize & score | 9–10        | Size/shape/intensity + overall confidence      |
| Review & export      | 11–12       | GUI plots + Markdown/JSON/PNG report           |

## 3. Two data paths (important)

**UM-BMID (.mat \_adi):** already cleaned; Module 2 should stay pass-through; tumor/healthy comes from metadata.

**UG Touchstone (.s2p):** full Module 2 Week 1–3 preprocessing is required.

## 4. Module status summary

| #  | Module                                      | Status                      |
|----|---------------------------------------------|-----------------------------|
| 1  | Data Acquisition and Validation             | Completed                   |
| 2  | Signal Preprocessing                        | Completed                   |
| 3  | Frequency-to-Time Conversion                | Completed                   |
| 4  | Initial Breast Image Reconstruction         | Completed                   |
| 5  | Quality Assessment and Beamformer Selection | Completed                   |
| 6  | Suspicious Region Localization (ROI)        | Completed                   |
| 7  | ROI High-Resolution Reconstruction          | Completed (core)            |
| 8  | Tumor Detection (Candidate Decision)        | Completed (core)            |
| 9  | Tumor Characterization                      | Completed (core)            |
| 10 | Reconstruction Confidence Assessment        | Completed (core)            |
| 11 | Visualization Dashboard                     | Completed (core)            |
| 12 | Automated Report Generation                 | Completed (current version) |

## 5. Detailed module descriptions

### Module 1 — Data Acquisition and Validation [Completed]

**Purpose:** Load measurement data into a common format and check that it is valid for imaging.

**What it does:**

- Loads MATLAB (.mat), UM-BMID multi-scan cubes, and Touchstone (.s2p / .sNp) files.
- Validates frequency vectors (ordering, spacing, finite values) and S-parameter integrity.
- Extracts metadata: antenna radius, tumor/healthy labels, tumor ground-truth position when present.
- Supports BMID scan selection (tumor vs healthy filter) in the GUI.

**What you see in the app / report:**

- Upload tab: file summary, frequency range, ports/traces.
- Session report section 1 (Dataset).

**Code:** `data_loader/`, `gui/upload_page.py`

### Module 2 — Signal Preprocessing [Completed]

**Purpose:** Clean and prepare S21 signals before reconstruction. Two data paths are supported.

**What it does:**

- UG .s2p track (full brief): parse S21, convert to complex linear, validate/sort, uniform grid, phase unwrap, spike fix (Hampel/median/local), optional averaging, mild filtering with NRMSE, Hamming, IFFT/time-domain, reference subtract, clutter removal, global normalize, Module-3 handoff export.
- UM-BMID \_adi track: use pass-through preset (filter/calib/artifact = none) because \_adi data is already reference-cleaned.
- Reports signal quality (SNR before/after) and Module 2 validation fields.

**What you see in the app / report:**

- Preprocessing tab plots (magnitude / phase).
- Session report section 2 + Module 2 validation + S21 magnitude/phase/IFFT figures.

**Code:** `preprocessing/` (`touchstone_s21`, `week3_time_domain`, `handover_export`, `pipeline`)

### Module 3 — Frequency-to-Time Conversion [Completed]

**Purpose:** Convert frequency-domain measurements into time-domain signals for beamforming.

**What it does:**

- Applies IFFT-based frequency-to-time conversion.
- Builds a consistent time axis from frequency sampling ( $dt = 1/(N df)$ ).
- Provides time-domain traces used by DAS / DMAS / DMAS-D4.

**What you see in the app / report:**

- IFFT time-domain figure in the report (when Week-3 / time path is available).
- Internal path used automatically during reconstruction.

**Code:** `reconstruction/iff.py`, `preprocessing/week3_time_domain.py`

## Module 4 — Initial Breast Image Reconstruction [Completed]

**Purpose:** Form a 2D microwave image of the breast / phantom from multi-antenna S21 data.

**What it does:**

- Runs three beamformers on the same geometry and FOV: DAS, DMAS, and DMAS-D4.
- Uses configurable antenna radius, arc, angle offset, flips, wave speed, and grid size.
- BMID-friendly defaults (e.g. 18 cm radius, 12 cm FOV options).

**What you see in the app / report:**

- Reconstruction Image tab (selected result).
- Comparison tab (three beamformers).
- Session report section 3 header + selected/beamformer PNGs.

**Code:** `reconstruction/das.py`, `dmass.py`, `dmass_d4.py`, `reconstruction_manager.py`

## Module 5 — Quality Assessment and Beamformer Selection [Completed]

**Purpose:** Score each reconstruction and choose the best image for ROI analysis.

**What it does:**

- Computes image metrics: SNR, SCR, CCR, contrast, FWHM, peak/mean.
- Ranks beamformers with selectable modes: quality auto, prefer DMAS-D4, force DMAS-D4, closest to tumor GT.
- Carries the selected image forward to Modules 6–10.

**What you see in the app / report:**

- Details → metrics / summary tables.
- Report: Beamformer quality scores + same-scale comparison figure.

**Code:** `quality/metrics.py`, `quality/beamformer_selector.py`

## Module 6 — Suspicious Region Localization (ROI) [Completed]

**Purpose:** Find the most suspicious hotspot on the selected reconstruction.

**What it does:**

- Detects ROI mask, bounding box, centroid, area, and score.
- Modes: peak score, prefer off-center, tight peak.
- Reports distance to tumor ground truth when BMID metadata is available.

**What you see in the app / report:**

- Selected image overlay / ROI box.
- Report: ROI subsection + localization error in validation table.

**Code:** `roi/roi_detector.py`

## Module 7 — ROI High-Resolution Reconstruction [Completed (core)]

**Purpose:** Zoom and re-image the ROI at higher resolution for clearer focus.

**What it does:**

- Reconstructs only the ROI bounding region on a finer grid.
- ROI scoring prefers compact bright foci and penalizes large edge-touching clutter.
- Produces the ROI refinement image used in reports and GUI.

**What you see in the app / report:**

- ROI Refinement view in the GUI.
- Report: ROI refinement subsection + `*_roi_refine.png`.

**Code:** `reconstruction/reconstruction_manager.py` (`reconstruct_high_resolution_roi`), `roi scoring`

## Module 8 — Tumor Detection (Candidate Decision) [Completed (core)]

**Purpose:** Decide whether the localized ROI looks tumor-like or clutter-like.

**What it does:**

- Builds suspicion features (off-center tendency, size, peak/mean, compactness).
- Applies penalties for large / edge-touching clutter regions.
- Outputs Yes/No candidate, confidence, suspicion score, threshold, and reason.

**What you see in the app / report:**

- Details → ROI table: Tumor Candidate / Confidence / Suspicion.
- Report: Tumor candidate decision subsection.

**Code:** `roi/roi_detector.py (classify_tumor_candidate)`

## Module 9 — Tumor Characterization [Completed (core)]

**Purpose:** Describe the candidate quantitatively in physical units.

**What it does:**

- Size: equivalent diameter, bbox width/height, area (cm / cm<sup>2</sup>).
- Shape: aspect ratio, compactness, eccentricity proxy.
- Intensity: peak, mean, peak-to-mean, local SCR, FWHM.
- Position: centroid (cm), radial offset from origin.

**What you see in the app / report:**

- Details → ROI characterization rows.
- Report: Tumor characterization (Module 9).

**Code:** `quality/characterization.py`

## Module 10 — Reconstruction Confidence Assessment [Completed (core)]

**Purpose:** Give one interpretable confidence score for the whole reconstruction decision.

**What it does:**

- Combines five components: image quality, detection, beamformer margin, localization, focus.
- Produces overall score (0–1) and label: low / medium / high.
- Adds short human-readable reasons for the score.

**What you see in the app / report:**

- Details → Overall Confidence.
- Report: Reconstruction confidence (Module 10) + quality-check notes.

**Code:** `quality/confidence.py`

## Module 11 — Visualization Dashboard [Completed (core)]

**Purpose:** Interactive GUI for the full workflow from load to result review.

**What it does:**

- Tabs for Upload, Preprocessing, Reconstruction (Image / Settings / Details / Comparison).
- Shows selected reconstruction, beamformer comparison, and ROI refinement.
- Light Fusion theme for readable controls on Windows.

**What you see in the app / report:**

- Desktop application windows and plots.
- PNG figures attached to the session report.

**Code:** `gui/main_window.py, gui/upload_page.py, gui/reconstruction_page.py, gui/styles.py`

## Module 12 — Automated Report Generation [Completed (current version)]

**Purpose:** Export a reviewable Markdown/JSON/PNG package of the full pipeline run.

**What it does:**

- Auto-saves latest\_session\_report.\* under project/results/ after preprocess/reconstruct.
- Manual Report button writes a timestamped package and can copy to another folder.
- Includes dataset, preprocessing, reconstruction, ROI, candidate, characterization, confidence, checks, next steps, and figures.

**What you see in the app / report:**

- results/latest\_session\_report.md (+ .json + PNGs).
- This Modules Overview PDF (guide document).

**Code:** `utils/session_report.py`, `gui/main_window.py`

## 6. What appears in the session report by module

| Module | Report content                                                                           |
|--------|------------------------------------------------------------------------------------------|
| 1      | Section 1 Dataset + metadata + frequency integrity checks                                |
| 2      | Section 2 Preprocessing settings, SNR, Module 2 validation, magnitude/phase/IFFT figures |
| 3–4    | Section 3 geometry + selected reconstruction image                                       |
| 5      | Beamformer quality scores + same-scale comparison figure                                 |
| 6      | ROI box/centroid + localization error vs tumor GT                                        |
| 7      | ROI refinement subsection + roi_refine PNG                                               |
| 8      | Tumor candidate decision (Yes/No, confidence, reason)                                    |
| 9      | Characterization metrics (cm size/shape/intensity)                                       |
| 10     | Overall confidence + component breakdown + notes                                         |
| 11     | All embedded PNG figures                                                                 |
| 12     | The report package itself (MD/JSON/PNG) under results/                                   |

## 7. Remaining polish (not missing modules)

- Calibrate Module 8–10 thresholds on a larger healthy-vs-tumor BMID subset.
- Further reduce edge/clutter ROI artifacts on difficult scans.
- Optional: use Week-3 time-domain handoff as default Module 3 input (currently filtered S21(f) is primary).
- Polish final report templates for guide submission formatting.

## 8. How to generate a run-specific session report

- Open the GUI (python main.py).
- Load data → run preprocessing → run reconstruction.
- Click Report (or use menu Download report).
- Files are written to project/results/ (latest\_session\_report.\*).